• Email: mikeethanh04@gmail.com • Github: github.com/mikeethanh

EDUCATION

UNIVERSITY OF TRANSPORT AND COMMUNICATIONS (UTC) **Hanoi, VietNam**
Bachelor of Science in Information Technology - International Training Program
Graduation Date: 6/2026
GPA: 3.3

TECHNICAL SKILLS

- **Program laguages:** Python, SQL
- **Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn, Sentence-Transformers
- **Frameworks:** PyTorch , FastAPI, LlamaIndex
- **Machine Learning Algorithms:** Linear Regression, Logistic Regression, KNN, Tree based model (Decision Tree, RandomForest,Ada boosting, Gradient Boosting , XGBoosting)
- **Deep Learning:** Neural Networks, CNN, RNN, LSTM, Transformer
- **LLMs: LLMs & NLP**: In-depth knowledge of Transformer-based models (BERT, LLaMA 3.2-8B).
- **MLOps & DevOps**: Docker, CI/CD (GitHub Actions), experiment tracking (Weights & Biases), cloud deployment (DigitalOcean Spaces, GPU droplets)
- **Others:**
  - Strong teamwork and collaboration skills, high sense of responsibility.
  - Continuous learning mindset, keeping up-to-date with the latest advancements in AI models.

PROJECTS

Vietnamese Legal Chatbot RAG System (AI/NLP Research & Development Project)

• **Link project**: https://github.com/mikeethanh/Vietnamese-Legal-Chatbot-RAG-System.git
• **Model**: https://huggingface.co/mikeethanh/vietnamese-legal-llama3.2-3b-merged-sft-v1
• **Architecture & MLOps**: Architected a production-grade Microservices-based system using FastAPI, Celery, và Redis to handle asynchronous tasks. Deployed with Docker and GitHub Actions (CI/CD) on DigitalOcean, served on an NVIDIA H100 GPU for serving, and monitored with Prometheus/Grafana.

• **Domain-Specific Fine-tuning**: Fine-tuned Llama-3.1-8B-Instruct on 100K Vietnamese legal examples using Unsloth (LoRA) on an NVIDIA H200 GPU. Achieved significant performance gains: reduced Average Perplexity from 7.92 to 2.63 and improved ROUGE-L F1 score by ~15% compared to the base model.

• **RAG Pipeline**: Engineered an advanced Agentic RAG using LlamaIndex with Hybrid Search (combining Qdrant vector search with BGE-M3 embeddings and BM25 keyword retrieval), optimized by Cohere rerank-multilingual-v3.0.

• **Data & Serving Integration**: Managed complex data flows: preprocessing from HuggingFace Datasets, version control via DigitalOcean Spaces, and real-time history tracking with MariaDB. Served model via OpenAI-compatible API.

Vietnamese Legal AI - LLaMA 3.2 Fine-tuning (Deep Learning Research Project)

Link project: https://github.com/mikeethanh/finetune_advance.git
• **Model**: https://huggingface.co/mikeethanh/vietnamese-legal-llama3.2-3b-merged-sft-v1
• **Dataset**: https://huggingface.co/datasets/mikeethanh/synthetic_legal_qa_3k
• **Core Innovation:** Developed a specialized Reasoning Model for Vietnamese Traffic Law by implementing

GRPO (Group Relative Policy Optimization), enabling the model to generate structured, step-by-step legal analysis before producing final citations.

• **Two-Stage Alignment Strategy:** Executed a robust training pipeline starting with Supervised Fine-Tuning (SFT) on a formatted dataset to establish base logic, followed by GRPO to align the model with human-centric legal reasoning memory efficiency on T4 GPUs (16GB VRAM).

• **Multi-objective Reward Engineering:** Designed a sophisticated reward system with a heavy emphasis on Linguistic Precision (Spelling/Grammar) and Format Adherence, effectively eliminating previous formatting inconsistencies and hallucinations in legal output.

• **High-Quality Data Pipeline:** Processed 98K+ legal samples (ViLQA, hoidap-tvpl) and augmented the dataset with 800+ complex traffic law scenarios verified through a Human-in-the-loop process to ensure 100% accuracy in reasoning paths.

• **Hardware-Aware Optimization:** Successfully optimized the Reinforcement Learning task to run on a single NVIDIA T4 (16GB) by implementing 4-bit Quantization and a strategic num_generations=3 configuration via the Unsloth framework, achieving a stable Training Loss of 0.57.

• **Result:** Transitioned the model from simple text generation to Structured Reasoning, utilizing <start_working_out> and <SOLUTION> tags to provide transparent, explainable, and legally-sound advice.

CERTIFICATIONS & AWARDS

- **TOEIC (4 Skills)**
  - Listening & Reading: 780 (Valid until Aug 23, 2025)
  - Speaking & Writing: 270 (Valid until Dec 17, 2026)
- **Third Prize - Scientific Research Competition**
  - Awarded for the research project "Supervision for Online Learning", which applied Deep Learning models for automated attendance tracking and student behavior analysis.
- **Samsung Algorithm and Application Program Certificate(SamSung)**
  - Achieved A+ in Samsung's competitive certification program, demonstrating strong problem-solving and algorithmic skills.

- **Coursera Certifications:**
  - Supervised Machine Learning: Regression and Classification (DeepLearning.AI – Coursera)
  - Advanced Learning Algorithms (DeepLearning.AI – Coursera)
  - Neural Networks and Deep Learning(DeepLearning.AI – Coursera)
  - Python for Data Science, AI & Development (IBM – Coursera)
- **Completed a module on MLOps2(Dataengineering) (fullstackdatascience.com)**
- **Fundamental of Agent (Hugging Face)**
- **Currently enrolled in an AI Agents course from Hugging Face.**
- **Currently enrolled in an AIOcourse(2024) from AIVIETNAM.**